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The Noise on Ultraviolet Astronomical Images Captured by Back-illuminated Silicon Sensors

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ABSTRACT

Ultraviolet (UV) astronomy currently relies on back-illuminated silicon imaging sensors. While the noise in these sensors is typically assumed to be dominated by photon shot noise, measurements from HST's Wide Field Camera 3 (WFC3) and the Interface Region Imaging Spectrograph (IRIS) reveal a significant discrepancy: the noise measured in the ultraviolet (UV) is systematically lower than theoretical predictions. We propose that this discrepancy is caused by partial charge collection (PCC), whereby a fraction of photogenerated electron-hole pairs recombine before they can be measured. We present a simple theoretical model, valid for wavelengths from 1 Å to 10 000 Å, that incorporates the effects of PCC and shows better agreement with the noise measurements from both WFC3 and IRIS, resolving the previously unexplained discrepancy. This finding implies that the signal-to-noise ratio achievable with these sensors is higher in the UV than previously understood, potentially impacting both future instrument proposals and the uncertainties of our current UV imagery. At about 2000 Å to 3500 Å, PCC is the dominant noise source in the high-signal limit.

Keywords: Astronomical instrumentation (799); Astronomical detectors (84); Ultraviolet astronomy (1736)

1. INTRODUCTION

Back-illuminated, silicon-based image sensors such as charge-coupled devices (CCDs) and complementary metal–oxide–semiconductor (CMOS) sensors are ubiquitous in ultraviolet (UV) astronomy, and are currently used in many UV instruments such as the Atmospheric Imaging Assembly (AIA) (J. R. Lemen et al. 2012), the Interface Region Imaging Spectrograph (IRIS) (B. De Pontieu et al. 2014), and HST's Wide Field Camera 3 (WFC3) (R. A. Kimble et al. 2008). Predicting the noise measured by these sensors is important both for quantifying the uncertainty on current astronomical observations as well as for evaluating the performance of proposed future instruments.

In the high-signal limit (that is, neglecting readout noise), the largest source of noise measured by these sensors is thought to be shot noise (R. A. Stern et al. 1986). This noise is due to the quantized nature of light (W. Schottky 1918) and is not a property of the sensor *per se*, but its magnitude depends on the number of measured photons, which is determined by the sensor's geometry and optical constants. Another well-known source of noise inherent to silicon sensors is Fano noise (U. Fano 1947), the random variation of the number of electrons produced per interacting photon. This noise source is small (D. Rodrigues et al. 2023), and often ignored entirely, but it can be combined with the shot noise to form the traditional estimate of the total noise (R. A. Stern et al. 1986; J. R. Janesick 2001).

This traditional model of shot noise and Fano noise is very accurate in the visible (M. Marinelli & J. Green 2024) and X-ray (S. Ishikawa et al. 2011; P. S. Athiray et al. 2020; B. D. Schwab et al. 2020) wavelength regimes. However, this model appears to be less accurate in the UV. For example, J. P. Wülser et al. (2018) showed that IRIS measured *less* noise in the far ultraviolet (FUV) channel than the traditional model would suggest and similar results were found in the near ultraviolet (NUV) by WFC3. T. Borders et al. (2010) measured the quantum yield of the WFC3 CCDs from narrowband flat fields between 208 nm and 400 nm and found it to be about 30% smaller than the theoretical prediction, which falls from 1.7 electrons per interacting photon at 200 nm to unity at 340 nm. The WFC3 instrument handbook, which summarizes this measurement, still describes its cause as unclear (M. Marinelli & J. Green 2024). This discrepancy indicates that the traditional model is incomplete, and that a more detailed model is needed to predict the noise measured by these sensors in the UV.

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In this work, we aim to resolve the UV noise discrepancy by developing a noise model which adds two additional effects to the traditional model. The first effect is partial charge collection (PCC), which occurs when electron-hole pairs recombine before they can be measured by the sensor (R. A. Stern et al. 1994; J. R. Janesick 2001). Noise due to PCC is a well-known effect, for example D. Rodrigues et al. (2023) accounted for PCC in their measurement of the Fano noise in soft X-ray (SXR), but its influence on the noise measured in UV has been rarely modeled in detail. The second effect is charge diffusion (or charge spreading), the tendency for photoelectrons to migrate into neighboring pixels (J. R. Janesick 2001), a mechanism suggested by both J. P. Wülser et al. (2018) and T. Borders et al. (2010) to explain the discrepancies they measured. Flat fields are already known to depart from Poisson statistics in the visible, where the variance is suppressed and reappears as a covariance between neighboring pixels, an effect attributed to the electrostatic repulsion of charge already collected and known as the brighter-fatter effect (A. Guyonnet et al. 2015; P. Astier et al. 2019). That mechanism is distinct from the one considered here. It distorts the pixel boundaries by an amount which grows with the accumulated signal, whereas charge diffusion displaces each electron independently of every other, and independently displacing the points of a Poisson process yields another Poisson process. Charge diffusion by itself therefore cannot alter the variance of a flat field at all. It becomes a noise term only when a single photon liberates several electrons which would otherwise be collected together, so it switches on with the quantum yield in the UV rather than with the signal level. The closest treatment of this combination that we are aware of is J. J. Givans et al. (2022), who extended a flat-field correlation formalism to include both quantum yield and charge diffusion for the infrared detectors of the Nancy Grace Roman Space Telescope. We will model the contribution of each of these effects to the total noise and compare our predicted noise to the noise measured by IRIS and WFC3. We will also provide noise predictions for AIA and the Multi-slit Solar Explorer (MUSE) (B. De Pontieu et al. 2020) which can be used for estimating the uncertainty of observations captured by these instruments. Finally, we have made available a reference implementation of our noise model in Python which is available to be installed from the Python Package Index (PyPI).

2. SENSOR MODEL

In this work, we will use the back-illuminated CCD model described in R. A. Stern et al. (1994) as a basis for our sensor model to make it directly comparable with previous works in the literature. Figure 1 is a schematic of our sensor model and it shows the light-sensitive region of the sensor to be an epitaxial silicon layer with a thickness D. A fraction of the incident photons are either absorbed in the oxide layer (thickness δ) or reflected at the first or second interface. The illuminated side of the epitaxial layer is considered to have a PCC region of thickness W, where electron-hole pairs can recombine before being measured, and there is a depletion region of thickness z_d on the non-illuminated side.

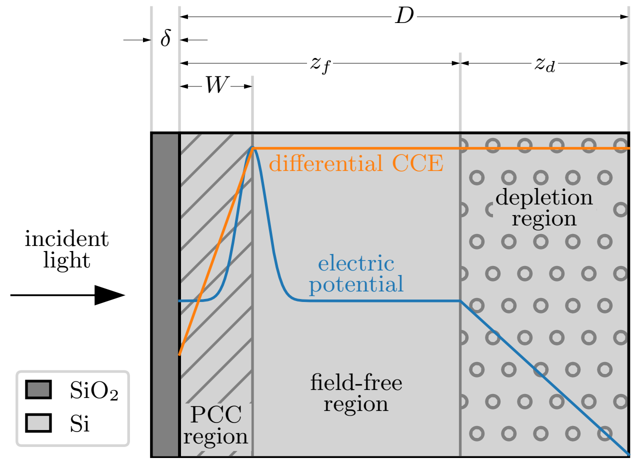

Figure 1. A schematic (not to scale) of the back-illuminated sensor model used in this work, labeled with the thicknesses of each layer. Overplotted is a *qualitative* example of the electric potential within the sensor that motivates the differential CCE.

In Section 2.1 we will calculate the signal predicted by this sensor model using the method described in R. A. Stern et al. (1994), adapted to more recent measurements. In Section 2.2 we will calculate the noise predicted by this sensor model in a self-consistent manner, and in Section 2.3 we will model the charge diffusion of this sensor which is needed for a complete description of the noise. Throughout this section we will adopt the convention that unprimed variables represent quantities incident on the sensor, primed variables represent quantities internal to the sensor, and the double-primed variables represent quantities measured by the sensor.

2.1. *Signal*

The average number of photoelectrons measured per incident photon is known as the quantum efficiency (QE). This is a common performance metric for measuring sensor sensitivity and it is given in J. R. Janesick (2001) as

$$\mathrm{QE}(\lambda, T) = \left\langle \frac{N_e''}{N_\gamma} \right\rangle = A(\lambda)\, \overline{n}(\lambda, T)\, \overline{\eta}(\lambda), \tag{1}$$

Table 1. The sensor model parameters which best fit the measurements in P. Boerner et al. (2012) and J. Heymes et al. (2020).

Source	implementation	η_0	W (nm)	δ (nm)	D (μm)	substrate roughness (nm)
P. Boerner et al. (2012)	`e2v_ccd203`	0.452	335	1.4	16	1.1
J. Heymes et al. (2020)	`e2v_ccd97`	0.334	153	4.6	14	4.8

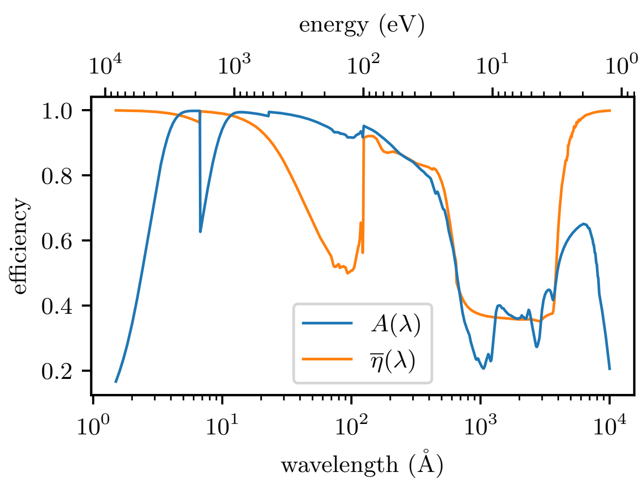

Figure 2. The fraction of incident light absorbed by the light-sensitive silicon layer and the CCE as a function of wavelength for the parameters inferred from the J. Heymes et al. (2020) measurements.

where λ is the wavelength of the incident photons in vacuum, T is the temperature of the sensor, $\langle X \rangle = \overline{X}$ denotes the expected value of a random variable X, N_e'' is the number of electrons measured by the sensor, N_γ is the total number of photons incident on the sensor, $A(\lambda)$ is the absorbance of the light-sensitive layer, $\overline{n}(\lambda, T)$ is the average quantum yield, the number of photoelectrons generated per absorbed photon, and $\overline{\eta}(\lambda)$ is the average charge-collection efficiency (CCE), the fraction of generated photoelectrons measured by the sensor.

The fraction of incident energy that is absorbed by the light sensitive layer is the absorbance, $A(\lambda)$, which is reduced by reflections at each interface, absorption in the oxide layer (UV), or by penetration through the device (SXR and infrared (IR)). $A(\lambda)$ can be determined from the optical constants of Si and SiO_2 using, for example, the popular IMD code (D. L. Windt 1998). For this work, we used our Python library, `optika` (R. T. Smart & C. C. Kankelborg 2024), which uses the transfer matrix method described in P. Yeh (1988) with the optical constants from E. D. Palik (1997), B. Henke et al. (1993), and L. V. R. de Marcos et al. (2016) to compute the electric field for every interface in the sensor. In Figure 2 we have plotted $A(\lambda)$ for the J. Heymes et al. (2020) parameters in Table 1.

In R. A. Stern et al. (1994), the authors assume no reflections from the unilluminated side of the sensor for simplicity. In this work, we compute the total change in Poynting flux into and out of the light-sensitive region of the sensor to determine $A(\lambda)$. This treatment introduces interference effects for infrared wavelengths, which can be seen on the right side of Figure 2.

Determining $\overline{n}(\lambda, T)$ over the entire wavelength range considered in this study is an area of ongoing research (G. W. Fraser et al. 1994; J. Geist 1996; F. Scholze et al. 1998; J. Fang et al. 2019, just to name a few). This work will use the quantum yield model developed by K. Ramanathan & N. Kurinsky (2020) which uses a phenomenological model of impact ionization to bridge the "UV gap", an area where direct measurements of the quantum yield are not yet available. They give the average quantum yield as

$$\overline{n}(E, T) = \begin{cases} \sum_{n=0}^{\infty} n p_n(E, T), & 0 < E \leq 50\,\mathrm{eV} \\ E/\epsilon_{eh}(T), & 50\,\mathrm{eV} < E < \infty, \end{cases} \tag{2}$$

where E is the energy of the incident photon, n is the actual quantum yield, $p_n(E, T)$ is a table of pair-creation probabilities which is provided in their supplementary material,

$$\epsilon_{eh}(T) = 1.7 E_g(T) + (0.084\,\mathrm{eV}^{-1}) A + 1.3\,\mathrm{eV}, \tag{3}$$

is the asymptotic mean energy per electron-hole pair, $A = 5.2\,\mathrm{eV}^2$, and

$$E_g(T) = 1.1692\,\mathrm{eV} - \frac{(4.9 \times 10^{-4}\,\mathrm{eV\,K^{-1}})\,T^2}{T + 655\,\mathrm{K}}, \tag{4}$$

is the bandgap energy of silicon.

In R. A. Stern et al. (1994), the average CCE is expressed in terms of a differential CCE, $\eta(z)$, which is the fraction of photoelectrons collected for a photon absorbed at a depth z into the epitaxial layer. The average CCE is then the first moment of the differential CCE weighted by the probability of absorbing a photon at a depth z,

$$\overline{\eta}(\lambda) = \frac{\int_0^\infty \eta(z)\, e^{-\alpha(\lambda) z}\, dz}{\int_0^\infty e^{-\alpha(\lambda) z}\, dz}, \tag{5}$$

where $\alpha(\lambda)$ is the absorption coefficient of silicon for the given wavelength.

In principle, $\eta(z)$ is a function of the exact implant profile which is usually impractical to measure, but see

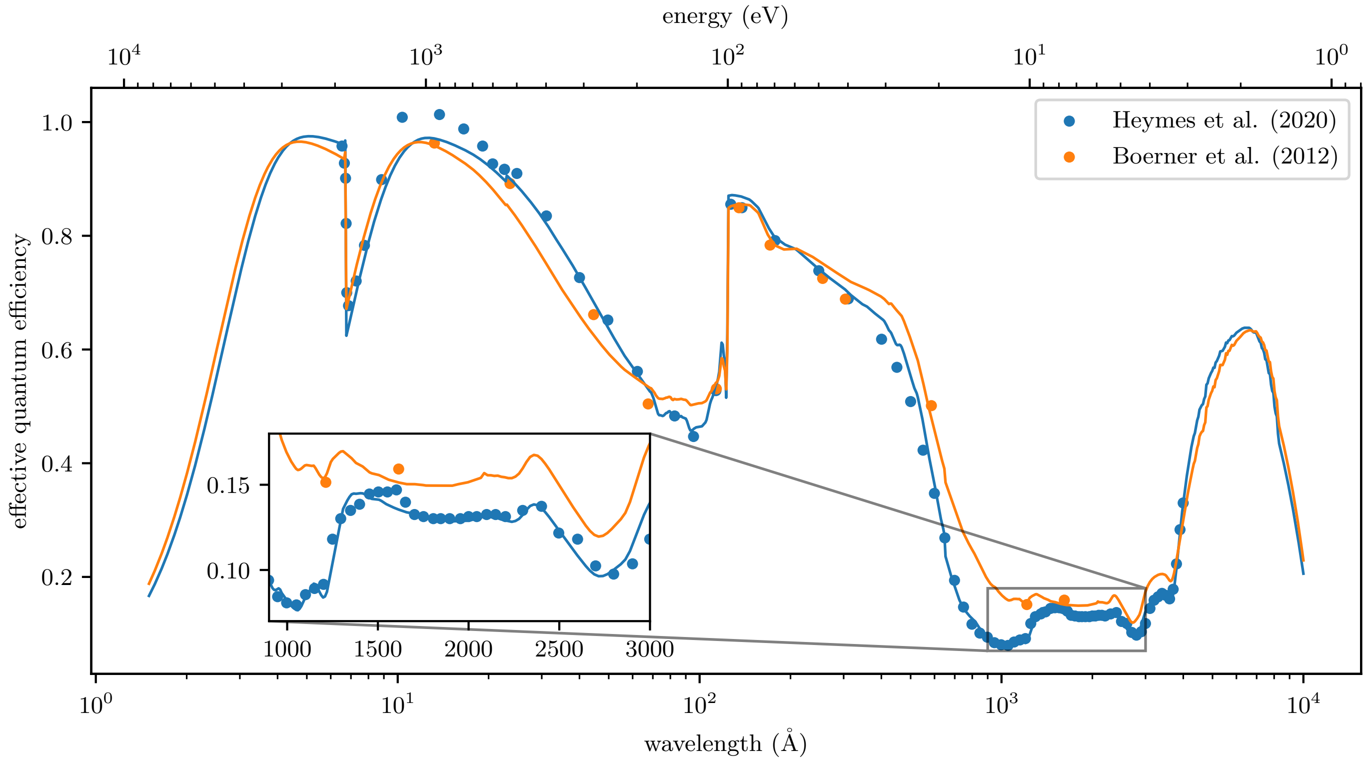

Figure 3. A comparison of EQE(λ) measured by P. Boerner et al. (2012) and J. Heymes et al. (2020), along with the best-fit models described in Table 1 plotted as lines with the same color as the data. The inset zooms into the ultraviolet to better visualize the difference between the measurement and model in this range.

R. A. Stern et al. (2004); P. Boerner et al. (2012) for a case where the authors did have a measurement of the implant profile provided by the manufacturer. In R. A. Stern et al. (1994), the authors instead adopt a piecewise-linear approximation of the differential CCE,

$$\eta(z) = \begin{cases} \eta_0 + (1-\eta_0)z/W, & 0 < z < W \\ 1, & W < z < \infty \end{cases} \tag{6}$$

where η_0 is the differential CCE at the back surface of the sensor. Plugging Equation 6 into Equation 5 yields an expression for the CCE,

$$\overline{\eta}(\lambda) = \eta_0 + \left(\frac{1-\eta_0}{\alpha W}\right)(1 - e^{-\alpha W}), \tag{7}$$

which can be used in Equation 1 to determine the QE. In Figure 2 we have plotted the average CCE for the J. Heymes et al. (2020) parameters in Table 1.

In R. A. Stern et al. (1994), the authors define an effective QE as

$$\mathrm{EQE}(\lambda) = A(\lambda)\,\overline{\eta}(\lambda), \tag{8}$$

which is found by taking the ratio of the current measured by the sensor to the current measured by a calibrated photodiode and is the quantity that is typically measured when calibrating an image sensor (R. A. Stern et al. 1994, 2004; P. Boerner et al. 2012). In Figure 3, we have plotted the measured, effective QE for two sources: P. Boerner et al. (2012) which measured the AIA CCDs at a few discrete wavelengths, and J. Heymes et al. (2020) which measured a Teledyne e2v CCD97 sensor over a wide wavelength range with high resolution using a monochromator.

Using the Nelder-Mead minimization algorithm (F. Gao & L. Han 2010), we found the free parameters of our model which best fit the data in P. Boerner et al. (2012) and J. Heymes et al. (2020). These models are plotted as solid lines in Figure 3, and the corresponding values of the free parameters are shown in Table 1. Throughout the remainder of this work we will use the model which best fits the J. Heymes et al. (2020) data.

2.2. *Noise*

Our noise model will consider three sources: shot noise from the quantized nature of the photons striking the sensor, Fano noise due to inherent randomness in the process which converts photons to electrons, and noise due to electrons randomly recombining before they can be measured by the sensor. This section will describe the statistics of each noise source and demonstrate a

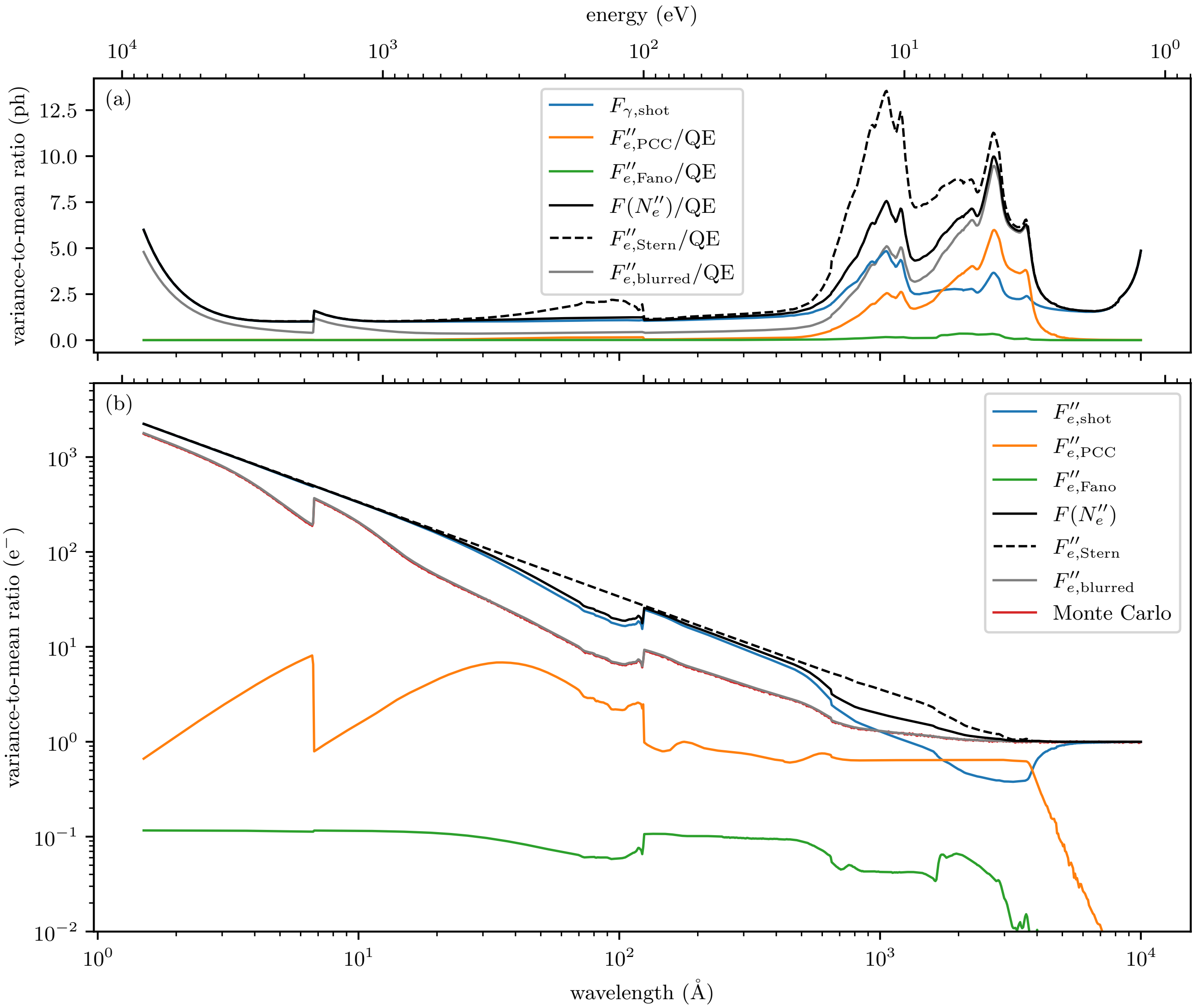

Figure 4. The total and component-wise VMR of our sensor model expressed in two different units: photons incident on the sensor (top) and electrons measured by the sensor (bottom). Plotted for comparison is the VMR of the R. A. Stern et al. (1986) noise model (dashed) and a Monte Carlo simulation of $F(N''_e)$ in red.

procedure which can simulate the noise measured by our model sensor for a given number of incident photons.

Throughout this work, we will measure noise in terms of a variance-to-mean ratio (VMR),

$$F(X) = \frac{\text{Var}(X)}{\overline{X}}, \tag{9}$$

where $\text{Var}(X)$ is the variance of X. Using the VMR instead of the signal-to-noise ratio (SNR) to express the noise is convenient since it is constant as a function of signal for the distributions studied here. For example, the VMR of a Poisson random variable is always unity since its variance and expectation value are equal. The VMR is not dimensionless (it has the same units as X), so we must take care to interpret the VMR in terms of the correct units. For a constant multiple of a random variable, aX,

$$F(aX) = aF(X). \tag{10}$$

We will use this relationship throughout this work to convert from incident photons to measured electrons and vice versa.

In Figure 4 we have plotted the VMR for the noise sources considered in this study in two different units: number of incident photons and number of measured electrons. Figure 4a is useful when *designing* an instrument since the number of photons incident on the sensor can be determined by the radiance of the source and the effective area of the instrument. Figure 4b is useful when *calibrating* an instrument since we directly measure these electrons. The details of the calculations for Figure 4 are presented in the subsections below.

2.2.1. *Shot Noise*

Photon shot noise is often assumed to be the leading noise contributor in UV astronomy (R. A. Stern et al. 1986; J. R. Lemen et al. 2012; B. De Pontieu et al. 2014). If the expected number of photons that interact with the light-sensitive layer is

$$\overline{N}'_\gamma = A\overline{N}_\gamma. \tag{11}$$

then the actual number of interacting photons, N'_γ, is sampled from the Poisson distribution,

$$N'_\gamma \leftarrow \text{Poisson}\big(\overline{N}'_\gamma\big). \tag{12}$$

As mentioned in the previous section, the VMR of this process is

$$F(N'_\gamma) = 1, \tag{13}$$

but this in terms of the number of absorbed photons, which is internal to the sensor. In terms of the expected number of incident photons given the number of absorbed photons the VMR is

$$F_{\gamma,\text{shot}} \equiv F\left(\frac{N'_\gamma}{A}\right) = \frac{1}{A} \tag{14}$$

since A is the conversion factor between incident photons and absorbed photons. $F_{\gamma,\text{shot}}$ is plotted in Figure 4a in blue and demonstrates that this expression is nearly unity across almost the entire wavelength range of the sensor except in the UV, where the absorbance is poor (Figure 2). We can also express the VMR of the shot noise in terms of the number of measured electrons by using the QE as the conversion factor between the unprimed and double-primed variables,

$$F''_{e,\text{shot}} = \overline{n}\,\overline{\eta}. \tag{15}$$

This equation is plotted in Figure 4b in blue, where we can see that it increases linearly with decreasing wavelength since the average quantum yield is increasing.

2.2.2. *Fano Noise*

The energy resolution of silicon detectors is ultimately limited due to Fano noise (U. Fano 1947), the random variation of the quantum yield n. Fano noise is usually expressed in terms of the Fano factor, $\mathcal{F} = F(n)$. Silicon is commonly accepted to have $\mathcal{F} \approx 0.1$ (J. R. Janesick 2001). In part due to variations of the Fano noise as a function of wavelength and temperature (G. W. Fraser et al. 1994), there is some disagreement in the literature around a more precise value for $\mathcal{F}$ (G. W. Fraser et al. 1994; B. Lowe 1997; M. N. Mazziotta 2008; I. V. Kotov et al. 2018; D. Rodrigues et al. 2021, 2023, & references therein). However, because the Fano noise is so small compared to the other noise sources considered in this study, the precise form of this noise model does not appreciably influence our conclusions.

As we mentioned in Section 2.1, this work uses the quantum yield model of K. Ramanathan & N. Kurinsky (2020). They provide the probability mass function (PMF) of the quantum yield distribution, $p_n(E,T)$, which we can use directly to generate random samples of the quantum yield for $E \leq 50\,\text{eV}$. For $E > 50\,\text{eV}$, we will assume that the quantum yield is a Gaussian distribution centered around $\overline{n}(\lambda, T)$, with standard deviation $\sqrt{\overline{n}(\lambda,T)\,\mathcal{F}(T)}$, where $\mathcal{F}(T)$ is the asymptotic Fano factor given by K. Ramanathan & N. Kurinsky (2020) as

$$\mathcal{F}(T) = (-0.028\,\text{eV}^{-1})\,E_g(T) + (0.0015\,\text{eV}^{-2})\,A + 0.14. \tag{16}$$

At 300 K, this corresponds to a Fano factor of approximately 0.114, which is generally consistent with the best-available measurement of the Fano factor by D. Rodrigues et al. (2021) at 6 keV.

$\mathcal{F}$ is a quantity internal to the sensor. The measured Fano factor is instead

$$F''_{e,\text{Fano}} = \overline{\eta}\mathcal{F} \tag{17}$$

since the conversion factor between the generated electrons and the measured electrons is the CCE. In Figure 4b we have plotted $F''_{e,\text{Fano}}$ in green. This shows that the Fano noise is the smallest contributor to the total noise across the entire wavelength range and is often at least an order of magnitude smaller than the other noise sources.

2.2.3. *Partial Charge Collection Noise*

In Figure 5, we have plotted the penetration depth in silicon vs. the thickness of the PCC region. We can see that there are two regions in the UV where the penetration depth is less than the thickness of the PCC region. In these regions, there is a significant chance that some of the electron-hole pairs will randomly recombine before being measured by the sensor. The stochastic nature of this process leads us to consider this as a new noise source, PCC noise.

In the R. A. Stern et al. (1994) charge-collection model, each photoelectron generated in the PCC region has a probability $\eta(z)$ of *not* recombining and subsequently being measured by the sensor. If n'_i is the quantum yield of the ith photon generated using the K. Ramanathan & N. Kurinsky (2020) model discussed in the

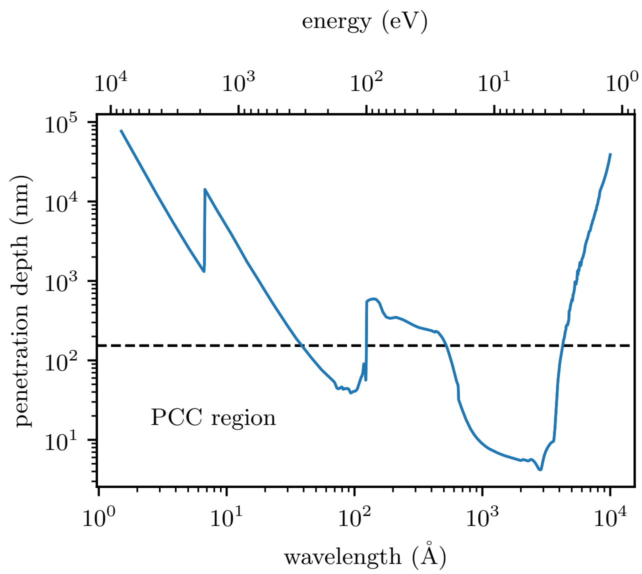

Figure 5. The penetration depth in silicon as a function of wavelength plotted against the depth of the PCC region.

previous section, we can express the *measured* quantum yield as

$$n_i'' \leftarrow \mathrm{Binomial}\big(n_i', \eta(z_i)\big), \tag{18}$$

where z_i is the penetration depth of the ith photon and $\mathrm{Binomial}(n, p)$ is a sample from the binomial distribution with n trials and probability of success p. We can use 18 to compute the total number of measured electrons given N_γ' absorbed photons as

$$N_e'' = \sum_{i=0}^{N_\gamma'} n_i''. \tag{19}$$

This expression is intended to be used in an instrument forward model to sample the distribution of measured electrons given an expected number of incident photons.

At first glance the VMR of this process looks intractable, since Equation 19 is a sum of binomial draws in which both the number of trials and the probability of success vary from one term to the next. It is made tractable by the observation made at the start of this section, that the VMR is constant as a function of signal, which leaves us free to evaluate it at whichever illumination is most convenient. The most convenient is the faintest. Consider an exposure so brief that the entire pixel array receives a single photon with probability ε and no photon at all otherwise. Let X be the number of electrons collected by a given pixel, and X_1 the number that same pixel collects on those occasions when the photon is present. The empty exposures contribute nothing to either average, so

$$\overline{X} = \varepsilon\,\overline{X_1}, \qquad \overline{X^2} = \varepsilon\,\overline{X_1^2}, \tag{20}$$

and the VMR of the exposure is

$$F(X) = \frac{\overline{X^2} - \overline{X}^2}{\overline{X}} = \frac{\overline{X_1^2}}{\overline{X_1}} - \varepsilon\,\overline{X_1}. \tag{21}$$

As the exposure is made fainter the second term vanishes, and the VMR is fixed entirely by the first two moments of what a single photon delivers to a single pixel. Since the VMR does not depend on the illumination, this limit is the answer at any illumination.

In this subsection charge diffusion is neglected, so the electrons liberated by that photon are all collected by the pixel it struck, and X_1 is simply the measured quantum yield n''. Both of its moments follow from Equation 18. For a photon which liberates n' electrons at a depth z, the measured yield n'' is binomial, with mean $n'\eta(z)$ and variance $n'\eta(z)\big(1-\eta(z)\big)$, so its second moment is

$$\overline{(n'')^2}\big|_{n',z} = n'\eta(1-\eta) + \big(n'\eta\big)^2. \tag{22}$$

Averaging over the absorption depth and the quantum yield, which are independent at a given wavelength, and using $\mathrm{Var}(n') = \mathcal{F}\overline{n}$ to evaluate $\overline{(n')^2}$, gives

$$\overline{(n'')^2} = \overline{n}\,\overline{\eta} + \overline{n}\big(\overline{n} + \mathcal{F} - 1\big)\overline{\eta^2}. \tag{23}$$

The mean is simply $\overline{n''} = \overline{n}\,\overline{\eta}$, so dividing Equation 23 by it gives the VMR of the electrons measured by the sensor,

$$F_e'' = 1 + \big(\overline{n} + \mathcal{F} - 1\big)\frac{\overline{\eta^2}}{\overline{\eta}}. \tag{24}$$

The two terms of Equation 23 are worth distinguishing, since they respond differently to charge diffusion in Section 2.3. Each factor of η is the probability that one electron survives recombination, so the number of factors counts the electrons a term describes. The first term carries a single factor and is the contribution of the electrons taken one at a time; it becomes the leading unity of Equation 24, which is why the VMR measured in electrons can never fall below one. The second term carries two factors, through $\overline{\eta^2}$, and describes the electrons two at a time. It is nonzero only because a single photon can liberate more than one electron, and it is the entire excess above the Poisson floor.

It is conventional to express this result in terms of the VMR of the CCE rather than its second moment. Taking the second moment of Equation 6 in a similar fashion to Equation 5 gives

$$F(\eta) = \frac{2e^{-\alpha W}}{\overline{\eta}}\left(\frac{1-\eta_0}{\alpha W}\right)^2\big(\sinh(\alpha W) - \alpha W\big), \tag{25}$$

and rearranging the definition of the VMR shows that the ratio appearing in Equation 24 is simply $\overline{\eta} + F(\eta)$.

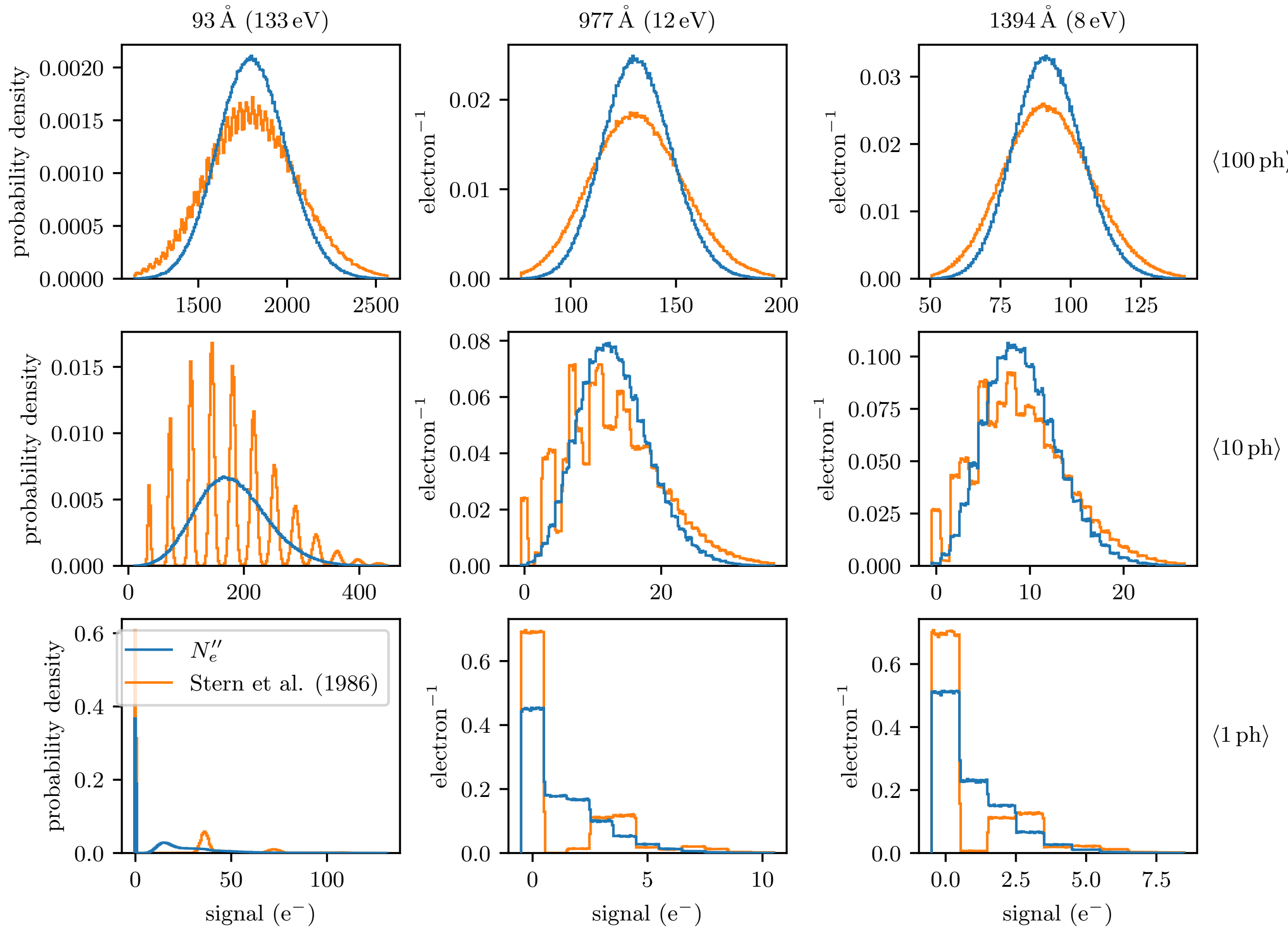

Figure 6. The probability distribution of the number of measured electrons for a given wavelength and expected number of *absorbed* photons calculated using 1000000 samples of Equation 19 and the R. A. Stern et al. (1986) noise model.

Substituting this and setting aside the shot noise of Equation 15 leaves the combined Fano and PCC contribution,

$$F''_{e,\text{sensor}} = F''_e - F''_{e,\text{shot}} = 1 + (\mathcal{F}-1)\overline{\eta} + \left(\overline{n} + \mathcal{F} - 1\right) F(\eta). \tag{26}$$

The contribution from only PCC noise is then

$$F''_{e,\text{PCC}} = F''_{e,\text{sensor}} - F''_{e,\text{Fano}}, \tag{27}$$

which is plotted in Figure 4 in orange. We can see that the PCC noise is usually very small compared to the shot noise, but in the UV from about 2000 Å to 3500 Å, it actually exceeds the shot noise and is the dominant contributor to the noise.

In Figure 6, we have plotted the PMFs of Equation 19 and the R. A. Stern et al. (1986) noise model to visualize the differences between these two distributions. Each model has been evaluated on a grid where each column of the grid represents a different wavelength (chosen to demonstrate the worst-case differences between the two models) and each row represents a different expected number of absorbed photons. When the number of absorbed photons is low, like in the bottom two rows of Figure 6, we can see that the R. A. Stern et al. (1986) model (orange) has a comb-like appearance caused by the Fano noise slightly blurring the PMF of the photon shot noise. In contrast, our model shows much less of this comb pattern since the PCC noise tends to blur the distribution further into a single peak. As the number of photons increases, both distributions tend towards a Gaussian, but our model remains narrower.

To ease adoption of this model, we've provided a reference implementation of Equation 19 in Python, `optika.sensors.signal()`, which is designed to be simple to use for existing and future instrument pipelines.

2.3. *Charge Diffusion*

In most back-illuminated imaging sensors used for UV astronomy, the depletion region (the region with significant electric field) does not extend all the way from the gate structure through the device. As a result, there is a so-called field-free region near the back of the sensor where photoelectrons must undergo a random walk to find their way to the depletion region where they can then be conducted to the terminals and measured (J. R. Janesick 2001). This random walk generally leads to a loss of spatial resolution measured by the sensor since electrons can diffuse to adjacent pixels. It can also lead to a reduction in the variance of a flat field measured by the sensor, but only where a single photon liberates more than one electron. The random walk displaces each electron independently of every other, so for a quantum yield of unity it carries one Poisson distribution of electrons into another and leaves the variance of a flat field unchanged. Where the quantum yield exceeds unity, the electrons liberated by a single photon would otherwise be collected together, and spreading them across neighboring pixels both reduces the variance and induces a covariance between those pixels.

Using Monte Carlo modeling, J. R. Janesick (2001) found the following expression for the standard deviation of the charge diffusion kernel:

$$\sigma(z) = \begin{cases} z_f\sqrt{1 - z/z_f}, & 0 < z < z_f \\ 0, & z_f < z < D, \end{cases} \tag{28}$$

where z is the distance from the back surface at which the photon is absorbed, $z_f = D - z_d$ is the thickness of the field-free region of the sensor, and z_d is the thickness of the depletion region. Using Equation 28, we can find the average variance of the charge diffusion kernel by taking a mean across the entire thickness of the sensor weighted by the probability of a photon being absorbed at that depth,

$$\overline{\sigma^2} = \frac{z_f\left(\alpha z_f + e^{-\alpha z_f} - 1\right)}{\alpha\left(1 - e^{-\alpha D}\right)}. \tag{29}$$

The thickness of the depletion region or the field-free region is difficult to measure, and depends on the voltage applied to the sensor and the charge collected at the terminals (R. A. Stern et al. 2004).

However, R. A. Stern et al. (2004) did estimate the size of the charge diffusion kernel, for two discrete wavelengths, of a 15 μm-thick (100 Ω-cm resistivity) CCD for the GOES Soft X-ray Imager. We can use these measurements to estimate the size of the depletion region and model the size of the charge diffusion kernel as a function of wavelength. R. A. Stern et al. (2004) did not directly measure the size of the charge diffusion kernel, instead they measured a quantity they named the mean charge capture (MCC), the fraction of charge captured by the central pixel. Naively, the MCC would be the integral of the charge diffusion kernel over the extent of a pixel. However, since a photon can strike anywhere within the central pixel, we need to convolve with a rectangle function the width of a pixel before integrating. If we assume that the charge diffusion kernel is a Gaussian with variance $\overline{\sigma^2}$, then the MCC is

$$m = \left[\frac{1}{\sqrt{\pi a}}\left(e^{-a} - 1\right) + \mathrm{erf}\left(\sqrt{a}\right)\right]^2, \tag{30}$$

where $a = d^2/2\overline{\sigma^2}$, d is the width of a pixel, and $\mathrm{erf}(x)$ is the error function.

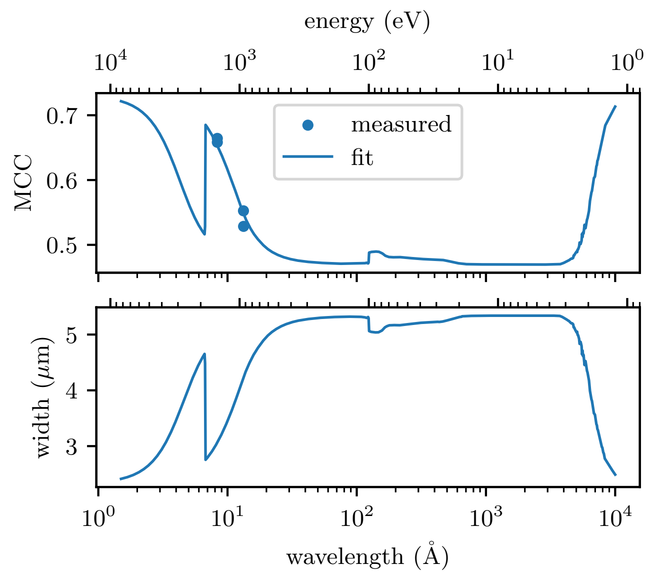

Figure 7. The top panel plots the MCC measured by R. A. Stern et al. (2004) and the fit of our model. The bottom panel shows the corresponding standard deviation of the charge diffusion kernel for the J. Heymes et al. (2020) sensor.

In the top panel of Figure 7, we have plotted a fit of Equation 30 to the measurements in R. A. Stern et al. (2004) which found $z_d = 8.7$ μm best matched the data. In the lower panel of Figure 7, we have plotted the corresponding standard deviation of the charge diffusion kernel as a function of wavelength which predicts that the charge diffusion is reasonably constant over much of the SXR and ultraviolet wavelengths since the penetration depth is low in this regime. Since the depletion thickness depends on the resistivity and bias voltage of a particular device, in Appendix A we check this model against the charge diffusion measured directly on the IRIS CCDs using the tracks left by energetic particles. The probability that two electrons liberated at the back surface are collected in the same pixel, which is the quantity that enters our noise model below, is 0.34 ± 0.01 on

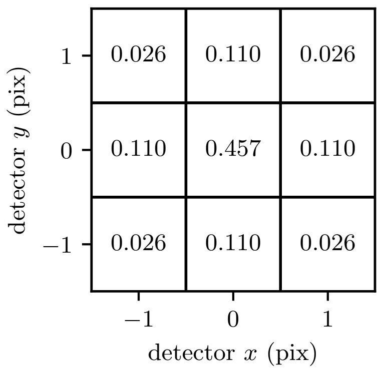

Figure 8. The charge diffusion kernel at 1400 Å convolved with a 13 μm IRIS pixel and integrated over the extent of each pixel.

the IRIS the Slit-Jaw Imager (SJI) CCD, compared to 0.34 predicted by this model.

To visualize the extent of this charge diffusion, we've included Figure 8, a 3×3 kernel where the value in each pixel has been computed in a manner similar to Equation 30, just with different limits of integration. We used the IRIS pixel size and a representative wavelength to demonstrate the worst-case scenario for IRIS. Convolving this kernel with an image approximates the spatial extent of the charge diffusion, but it does not describe the process faithfully. A convolution assigns a fractional number of electrons to each pixel, whereas every electron is in fact collected by exactly one pixel, and it treats the displacement of each electron as independent, whereas the electrons liberated by a single photon share both an absorption depth and a sub-pixel origin. We therefore diffuse each electron individually, displacing it by $\mathcal{N}\big(0, \sigma^2(z_i)\big)$ in each direction and rounding to the nearest pixel, which conserves charge exactly and preserves the correlation between electrons from the same photon. This is implemented by `optika.sensors.signal()`, and the kernel in Figure 8 was computed with `optika.sensors.kernel_diffusion()`, which can be used to visualize the charge diffusion for other wavelengths and pixel sizes.

Charge diffusion does not add a noise source of its own. It moves electrons between pixels without creating or destroying any, so it leaves the mean of a flat-field image unchanged, and it reduces the variance only by weakening the correlation between electrons that came from the same photon. To find its effect on the noise we return to the faint exposure of Section 2.2, in which the array receives at most one photon. The image is then nothing but the charge liberated by that photon, and the only thing diffusion changes is how it is divided between the pixels, so we now ask how many of the photon's electrons each *pixel* collects rather than how many survive in total. Let m be the number of electrons which survive recombination, as in Section 2.2, and let q_j be the probability that any one of *those* electrons is collected by pixel j. Recombination has therefore already been accounted for in m, and q_j describes only where the surviving charge is delivered, so that $\sum_j q_j = 1$; the CCE enters this section through m and not through q_j. The electrons are displaced independently of one another, so the photon contributes $X_j \leftarrow \text{Binomial}(m, q_j)$ to pixel j, and the second moment summed over the pixels is

$$\sum_j \overline{X_j^2} = m \sum_j q_j - m \sum_j q_j^2 + m^2 \sum_j q_j^2 = m + m(m-1) \sum_j q_j^2. \tag{31}$$

Two quantities have appeared which were not present in Section 2.2. The sum $\sum_j q_j^2$ is the probability that two electrons displaced independently are collected by the same pixel, and the factor $m(m-1)$ multiplying it is the number of ordered pairs of distinct electrons the photon delivered. Neither was introduced by hand. Both are produced by the algebra, and they arrive together, because the only way that spreading the charge can alter the second moment is by separating electrons which would otherwise have been counted in the same pixel. Where no charge is shared, $\sum_j q_j^2 = 1$, Equation 31 collapses to m^2, and the result of Section 2.2 is recovered; this is why the pairs are invisible there and unavoidable here. The opposite extreme is equally simple. Where the charge is spread so widely that no two electrons share a pixel, $\sum_j q_j^2$ falls to zero and the sum of squares collapses to m, which is to say the image becomes a set of electrons bearing no relation to one another and the VMR falls to unity. Between these limits the VMR measured in electrons is nothing more than the ratio of the sum of the squares of a single-photon image to the sum of that same image, and charge diffusion moves it from one end of that range to the other.

It remains to evaluate $\sum_j q_j^2$ for the charge diffusion described by Equation 28. Two electrons liberated by the same photon begin their random walks from a common absorption depth and a common position within a pixel, so how often they are collected together depends on how far the charge spreads at the depth where the photon was absorbed, expressed in units of the pixel width,

$$\begin{aligned} \mathcal{P}(z) &= p\big(\sigma(z)/d\big)^2, \\ p(s) &= \operatorname{erf}\left(\frac{1}{2s}\right) - \frac{2s}{\sqrt{\pi}}\left(1 - e^{-1/4s^2}\right), \end{aligned} \tag{32}$$

where $\mathcal{P}(z)$ is $\sum_j q_j^2$ averaged over the position of the photon within its pixel, $p(s)$ is the corresponding probability for a single axis, d is the width of a pixel, and the square accounts for the two directions across the face of the sensor, which spread independently of one another. In the limit of a narrow charge cloud this probability approaches unity and the electrons remain together; in the opposite limit it falls to zero and they are scattered independently.

Averaging Equation 31 over the quantum yield and the absorption depth, exactly as in Section 2.2, requires $\overline{m} = \overline{n}\,\overline{\eta}$ as before, and $\overline{m(m-1)} = \overline{n}\big(\overline{n}+\mathcal{F}-1\big)\overline{\eta^2}$, since a pair survives recombination only if both of its electrons do. Dividing by the mean gives the VMR measured by a sensor with charge diffusion,

$$F''_{e,\mathrm{diff}} = 1 + \big(\overline{n} + \mathcal{F} - 1\big)\frac{\overline{\mathcal{P}\eta^2}}{\overline{\eta}}. \tag{33}$$

Comparing with Equation 24, the effect of charge diffusion is to discount the two-electron term by the probability that the two electrons are in fact measured together, and to leave the one-electron term untouched. The discount is applied inside the average over absorption depth rather than outside it, since the distance the charge spreads and the fraction of it that survives recombination both depend on where in the sensor the photon was absorbed, and the two are correlated: photons absorbed near the back surface are the ones which both recombine most readily and diffuse the furthest. Equation 33 is the form implemented by `optika.sensors.vmr_signal()` and plotted for IRIS in Figure 4. It assumes the illumination is uniform, so that the charge leaving each pixel is balanced on average by the charge arriving from its neighbors; near the edge of the sensor, where diffused charge is lost, it is an approximation.

3. RESULTS AND DISCUSSION

Since $\mathrm{EQE}(\lambda)$ can be interpreted as an efficiency, it is tempting to think that the expected number of measured photons and their variance is $\mathrm{EQE} \times \overline{N}_\gamma$. However, using the effective QE in this way is equivalent to an all-or-nothing charge collection model where either all the electrons associated with an absorbed photon are measured or none of them are. This simple noise model is formalized in Equations 9 and 10 of R. A. Stern et al. (1986), which gives the VMR of the number of measured electrons as

$$F''_{e,\mathrm{Stern}} = \overline{n} + \mathcal{F} \tag{34}$$

This is the noise model used by T. Borders et al. (2010) and J. P. Wülser et al. (2018). For comparison, the VMR predicted by our undiffused noise model is the sum of $F''_{e,\mathrm{shot}}$ and $F''_{e,\mathrm{sensor}}$,

$$F(N''_e) = 1 + \big(\overline{n} + \mathcal{F} - 1\big)\big(\overline{\eta} + F(\eta)\big). \tag{35}$$

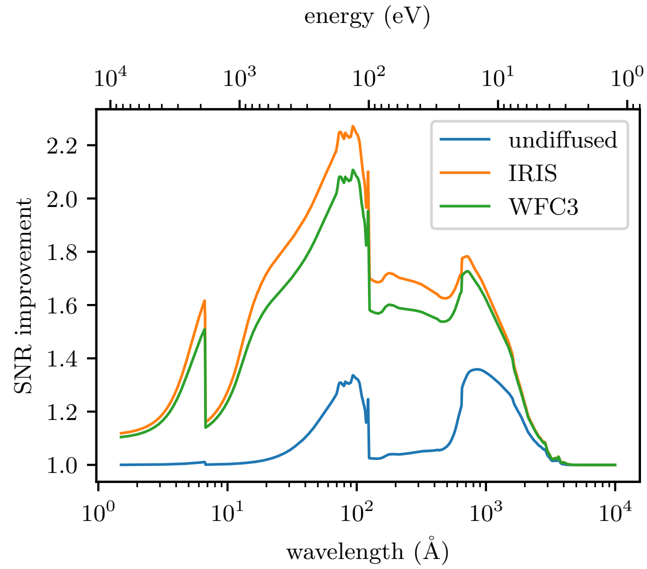

Figure 9. The SNR improvement predicted by our noise model compared to the R. A. Stern et al. (1986) model. In blue we have plotted the SNR improvement ignoring charge diffusion. In orange and green we have plotted the SNR improvement, including charge diffusion, for a flat-field image observed by IRIS and WFC3.

In Figure 4, we have compared $F''_{e,\mathrm{Stern}}$ (dashed) to our undiffused model, $F(N''_e)$ (black). Over most of the SXR and visible wavelengths the R. A. Stern et al. (1986) model is a good approximation of the undiffused model developed in this work since the penetration depth is deeper than the PCC region. However, in the UV, where the penetration depth is very shallow, R. A. Stern et al. (1986) overestimates the noise compared to our undiffused model.

It may seem paradoxical to add a noise source and then measure less total noise, but this is because more photons are measured in our model than the all-or-nothing model. Even though PCC introduces noise, it is less noise than there would be if the photon was undetected. This is good news for designers of UV astronomical instruments since there is less noise than expected from the R. A. Stern et al. (1986) model in this wavelength range.

In Figure 9, we have plotted the ratio of the SNR predicted by our model to the SNR predicted by the traditional model. The blue line is the improvement ratio for the undiffused model and it shows that there are two regions, one from 30-100 Å and another from 500-2000 Å, where the noise statistics deviate from the traditional model, in some cases by up to ~30%. This figure can be used to quickly estimate the importance of PCC noise in a given wavelength range.

Table 2. The VMR predicted by our model (including charge diffusion) for prominent wavelengths in selected UV observatories, in both incident photon and measured electron units. Also included is the SNR improvement ratio between our model and the traditional noise model as well as the standard deviation of the charge diffusion kernel.

Instrument	wavelength (Å)	VMR (ph)	VMR (e^-)	SNR improvement	diffusion width(μm)
AIA	94	0.43	6.96	2.27	5.32
	131	0.40	8.98	1.69	5.05
	171	0.42	6.72	1.72	5.15
	193	0.44	5.96	1.71	5.17
	211	0.45	5.53	1.70	5.17
	304	0.51	3.96	1.68	5.21
	335	0.53	3.64	1.67	5.21
	1600	3.83	1.15	1.42	5.34
	1700	4.51	1.12	1.35	5.34
IRIS	1330	3.22	1.20	1.52	5.34
	1400	3.23	1.19	1.49	5.34
	2796	9.21	1.02	1.08	5.34
	2832	8.93	1.02	1.09	5.34
MUSE	108	0.43	6.75	2.15	5.32
	171	0.42	6.72	1.72	5.15
	284	0.49	4.21	1.69	5.20
WFC3	2080	6.08	1.07	1.19	5.34
	2240	6.57	1.05	1.15	5.34
	2400	6.40	1.04	1.13	5.34
	2560	7.90	1.03	1.10	5.34
	2720	9.51	1.03	1.09	5.34
	2880	8.51	1.03	1.08	5.34
	3040	6.56	1.02	1.05	5.34
	4000	3.01	1.01	1.01	5.30

In Table 2, we have calculated the VMR in terms of incident photons and measured electrons for the target wavelengths of a few popular and upcoming solar instruments: AIA (J. R. Lemen et al. 2012), IRIS (B. De Pontieu et al. 2014), and MUSE (B. De Pontieu et al. 2020). The results show that for the extreme ultraviolet (EUV) channels, AIA and MUSE are nearly shot-noise-limited since the VMR in units of incident photons is near unity. Tabulated in the second-to-last column of Table 2 is the improvement factor between the VMR of the R. A. Stern et al. (1986) noise model and our noise model. These ratios show that the AIA 94 Å and 1600 Å, IRIS 1330 Å and 1400 Å, and MUSE 108 Å channels are predicted to have at least 20% more SNR than the traditional noise model would suggest. These results include the influence of charge diffusion, and we've also tabulated the size of the charge diffusion kernel in the last column, since it sets the scale over which the charge from a single photon is shared between neighboring pixels.

In Table 3, we've attempted to reproduce the measurements of J. P. Wülser et al. (2018) and T. Borders et al. (2010) by taking the ratio of the VMR of a simulated UV flat-field image to the VMR of a simulated visible flat-field image. The flat-field images were created by drawing samples from Equation 19 and then diffusing each measured electron individually, using the pixel size of the corresponding instrument. This table shows that the discrepancy discussed in the introduction is substantially reduced, though our model does not reproduce the measurements exactly. For WFC3, T. Borders et al. (2010) measured 1.09 at 2080 Å where the traditional model expects about 1.7, and our model predicts 1.06, close to the measured value. For IRIS, J. P. Wülser et al. (2018) measured 1.5 at 1350 Å where the traditional model expects about 2, and our model predicts

Table 3. The ratio of the VMR of a UV flat-field image to the VMR of a visible-light flat-field image. The second column from the right is the value of this ratio measured by T. Borders et al. (2010) and J. P. Wülser et al. (2018). For IRIS this is the ratio of the inverse camera gains measured from photon-transfer curves, 12 photons per data number using a deuterium lamp at 1350 Å and 18 photons per data number using the onboard blue light-emitting diode (J. P. Wülser et al. 2018), which we take to be the 430 nm device flown on the GOES Soft X-ray Imager (R. A. Stern et al. 2004). The rightmost column is the value of this quantity predicted by our noise model (including charge diffusion).

Instrument	λ_{UV} (Å)	λ_{vis} (Å)	measurement	model
IRIS	1350	4300	1.50	1.19
WFC3	2080	4000	1.09	1.06
	2240	4000	1.09	1.05
	2400	4000	1.07	1.04
	2560	4000	1.04	1.03
	2720	4000	1.04	1.02
	2880	4000	1.03	1.02
	3040	4000	1.01	1.01

1.19. This is much nearer the measurement than the traditional model, but it underestimates it, where the traditional model overestimates it. Our model therefore accounts for most of the reported discrepancy in both cases, but it overcorrects for IRIS. Evaluating the same model without charge diffusion raises the predicted IRIS ratio to approximately the measured value, so the disagreement could be explained if we had overestimated the charge diffusion of the IRIS CCDs. There are two reasons to suspect that we might have. The thickness of the depletion region was fit to the measurements of R. A. Stern et al. (2004), which were made on a different sensor operated at a different voltage, and the depletion thickness depends on both the resistivity of the silicon and the applied bias. Furthermore, we model the charge cloud as a Gaussian, whereas G. G. Pavlov & J. A. Nousek (1999) show that the true radial distribution is peaked more strongly and has heavier tails, which would place more of the charge liberated by a single photon in a single pixel than we have assumed. To test this, we measured the charge diffusion of the IRIS CCDs directly using the tracks left by energetic particles (Appendix A). On the SJI CCD, the sensor on which J. P. Wülser et al. (2018) measured the photon-transfer curves, the probability that two electrons liberated at the back surface are collected in the same pixel is 0.34 ± 0.01, compared to 0.34 for our model. This probability is measured without assuming a shape for the kernel and is the quantity which enters Equation 33, so neither the depletion thickness nor the shape of the kernel can account for the disagreement.

With the charge diffusion fixed by measurement, the measured ratio constrains the remaining ingredients of Equation 33: the quantum yield, which is well established at 1350 Å, and the CCE. Reproducing the measured ratio would require the CCE at the back surface to be closer to unity than the 0.334 implied by the QE measurements of J. Heymes et al. (2020), so that nearly every photon liberates a full electron and the PCC contributes little to the variance. Alternatively, the measurement itself may be biased. The photon-transfer curve at 1350 Å was measured with a deuterium lamp, whose illumination is less uniform and less stable than that of the visible light-emitting diode used for the other curves, and any spatial or temporal nonuniformity in the illumination adds variance which a photon-transfer curve interprets as a larger VMR. We cannot distinguish between these possibilities with the published data, and a photon-transfer measurement with a monochromatic, spatially uniform UV source on a sensor whose CCE is independently known would settle the question.

4. CONCLUSIONS AND FUTURE WORK

This work aims to resolve the discrepancy identified by T. Borders et al. (2010) and J. P. Wülser et al. (2018) in the predicted vs. observed noise measured by back-illuminated silicon imaging sensors. To accomplish this, we reconsidered the effects of PCC and charge diffusion on the noise by developing a simple model of a back-illuminated sensor based on the CCE introduced by R. A. Stern et al. (1994) and a charge diffusion model found by J. R. Janesick (2001). By analyzing these effects in a self-consistent manner, we were able to mostly resolve the discrepancy, and we propose that PCC is the most important effect to consider in the UV. The width of the charge diffusion kernel is nearly independent of wavelength, but its effect on the VMR is not, since diffusion suppresses the photon-correlated term, which grows with the quantum yield.

Our sensor model predicts that there are two UV bands, 30-100 Å and 500-2000 Å, where PCC effects should be considered for a correct noise estimate. We evaluated our sensor model for some popular and upcoming UV astronomical instruments and found that there are a few channels in each instrument where our model implies that the SNR is better than the traditional model would suggest. We also have provided a reference implementation of our sensor model in Python, `optika.sensors`, to make this noise model simple to integrate with existing instrument data processing pipelines.

This work does not consider read noise or other noise sources since those will be specific to the particular camera electronics of each instrument, and also cannot be characterized using a VMR. Instead, this work focuses only on the noise intrinsic to the charge-generation process, since this is comparatively consistent across different instruments.

We plan to use this work to model the noise for the EUV Snapshot Imaging Spectrograph (ESIS) (J. D. Parker et al. 2022), as well as the Full-sun Ultraviolet Rocket Spectrometer (FURST), sounding-rocket-based spectrographs developed by our research group for observing the solar atmosphere.

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NASA grants XXXX, XXXX and XXXX. This article is an example of an executable paper (J. Lasser 2020), all of the simulations, figures, and tables are generated dynamically when this article is created. This helps prevent mistakes and allows this research to be as repeatable as possible. The code to create this document is available at https://github.com/roytsmart/ccd-noise-paper.

APPENDIX

A. MEASURING THE CHARGE DIFFUSION KERNEL WITH PARTICLE TRACKS

The charge diffusion model of Section 2.3 rests on a depletion thickness fit to the MCC measurements of R. A. Stern et al. (2004), which were made on a different sensor operated at a different voltage. Here we measure the charge diffusion of the IRIS CCDs themselves using the tracks left by energetic particles.

A charged particle crossing the sensor at glancing incidence deposits charge along a straight line running from the back surface to the front surface. Each row (or column) of pixels crossed by the track therefore samples the charge cloud at a different, known depth, and the spread of the charge across the track in that row measures the width of the charge diffusion kernel at that depth. Charge deposited in the field-free region diffuses in the same way regardless of how it was liberated, so the tracks measure the same kernel that applies to photoelectrons.

We searched 3249 level-1 frames from the IRIS FUV spectrograph and the SJI for tracks (Table 4), favoring frames taken while the spacecraft crossed the the South Atlantic Anomaly (SAA), where the flux of energetic protons is greatest, and frames in which the slit or the field of view was at least partly off the solar limb, so that the tracks could be found against a dark background. For the SJI frames only the off-limb half of the field of view was used. The background of each frame was estimated as the trimmed mean of the frames in the same observing sequence and subtracted, and the read noise was estimated from the median absolute deviation of the residual. Pixels more than five times the read noise above the background were labeled, connected groups of labeled pixels were fit with a straight line, and we kept the groups spanning at least twelve rows (or columns) with a slope of less than 0.35 pixels per row, so that each track is nearly aligned with the pixel grid and its cross-section is sampled once per row. A seven-pixel-wide cutout centered on the fitted line was extracted for each track, keeping the slices with more than 240 electrons, which yielded 1070 tracks.

For each track we modeled the charge in every slice as spread uniformly along the fitted centerline and diffused as a Gaussian with standard deviation

$$\sigma(t) = \sigma_{\max}\sqrt{1 - t/t_c}, \quad t < t_c, \tag{A1}$$

and zero otherwise, where $t = z/D$ is the fractional depth of the slice below the back surface, $t_c = z_f/D$ is the fractional thickness of the field-free region, and $\sigma_{\max}$ is the width of the charge cloud at the back surface. This is Equation 28 rewritten in terms of the fractional depth, since we observe the length of the track in slices rather than the thickness of the sensor in microns. A track that enters the back surface and exits the front surface spans the full thickness of the sensor, so the fractional depth of slice i of an N-slice track is $t = (i + 1/2)/N$, up to the orientation of the track, which we fit along with t_c and $\sigma_{\max}$ by exhaustive search on a grid of 21 values in each of $t_c \in [0, 1]$ and $\sigma_{\max} \in [0, 10]$ μm. At each grid point the offset and tilt of the centerline are adjusted to minimize a robust misfit, $\sum \ln(1 + r^2/2)$, where r is the residual of the fraction of each slice's charge in each pixel in units of the read noise.

Table 4. The charge diffusion measured from glancing particle tracks on the three IRIS CCDs. The number of frames is the number of level-1 images searched, with the number taken inside the SAA in parentheses. The flat tracks are those which constrain t_c to within 0.15 and show no Bragg rise along their length. t_c and $\sigma_{\max}$ are the medians of the per-track fits, with the interquartile range in brackets. p is the probability that two electrons deposited within $D/10$ of the back surface are collected in the same column, $\mathcal{P} = p^2$ is the corresponding probability that they are collected in the same pixel, and the last column is $\mathcal{P}$ predicted by the model used in this work for the same tracks.

CCD	frames	tracks	flat	t_c	$\sigma_{\max}$ (μm)	p	$\mathcal{P}$	$\mathcal{P}_{\text{model}}$
FUV1	2091 (211)	298	106	0.45 [0.35, 0.65]	4.5 [3.0, 5.5]	0.647 ± 0.025	0.419 ± 0.033	0.325
FUV2	2091 (211)	451	171	0.45 [0.40, 0.55]	5.5 [5.0, 6.5]	0.577 ± 0.012	0.332 ± 0.013	0.331
SJI	1158 (469)	321	140	0.45 [0.40, 0.50]	5.5 [5.0, 6.0]	0.581 ± 0.009	0.338 ± 0.010	0.340

The useful tracks are those which cross the full thickness of the sensor at nearly constant energy loss. We therefore kept only the tracks for which the fit constrains t_c to within 0.15 and improves on a model with no diffusion by at least ten units of misfit, and for which the median charge per slice in the last third of the track is within 50% of that in the first third, since a rise in the deposited charge along the track (a Bragg peak) indicates that the particle stopped inside the sensor and did not cross its full thickness. We call these the flat tracks; 417 tracks pass these cuts, 140 of them on the SJI CCD.

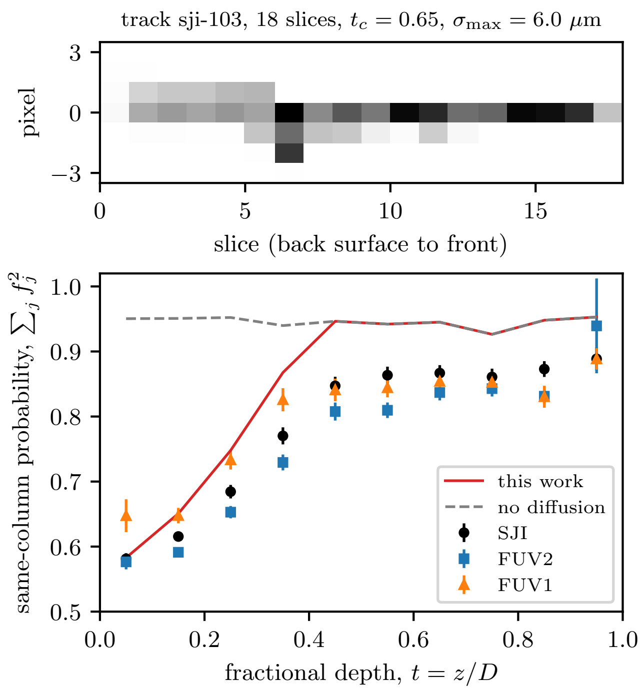

Figure 10. Top: a glancing particle track on the IRIS SJI CCD, in thousands of electrons per pixel, rotated so that it runs from the back surface (left) to the front surface (right). The charge is spread over several columns where the particle is near the back surface and is confined to one column where it is deep in the depletion region. Bottom: the probability that two electrons deposited in the same slice are collected in the same column, averaged over the flat tracks on each CCD in bins of fractional depth, with the standard error of each mean. The solid line is the prediction of the model used in this work (Equation 28 with $z_d = 8.7$ μm), evaluated at the fitted centerline of each SJI track and averaged in the same bins, and the dashed line is the same with no charge diffusion.

Figure 10 shows one of the flat tracks along with the probability that two electrons deposited in the same slice are collected in the same column, $\sum_j f_j^2$ where f_j is the fraction of the slice's charge in column j, averaged over the flat tracks on each CCD in bins of fractional depth. The bias from the read noise, $\sum_j \epsilon_j^2$ where ϵ_j is the read noise in units of the charge in the slice, has been subtracted. This quantity requires no model of the shape of the kernel, and its value near the back surface is the one-dimensional analogue of Equation 32, the quantity which enters our noise model through Equation 33. Since the kernel is separable, the probability that two electrons are collected in the same pixel is the square of the probability that they are collected in the same column. Table 4 lists the medians of the fitted parameters and the same-column and same-pixel probabilities for the slices within $D/10$ of the back surface, alongside the same-pixel probability predicted by our model evaluated along the same tracks.

On the SJI CCD, the sensor on which J. P. Wülser et al. (2018) measured the photon-transfer curves that we compare to in Section 3, the probability that two electrons liberated at the back surface are collected in the same pixel is 0.34 ± 0.01, in agreement with the 0.34 predicted by Equation 28 with $z_d = 8.7\ \mu$m. The FUV2 CCD is indistinguishable from the SJI CCD, while the FUV1 CCD appears to spread its charge over fewer pixels. The depletion thickness depends on the resistivity of the wafer and on the applied bias, both of which can differ between devices, but the FUV1 CCD is not the sensor to which we compare our model.

Two features of Figure 10 deserve comment. First, the shallowest bin averages over depths from the back surface to $D/10$, so the comparison in Table 4 is between averages over that bin rather than at the surface itself, and the model is averaged in the same way. Second, the fitted t_c, with a median of 0.45 on the SJI CCD, is somewhat larger than the 0.38 of our model, and beyond t_c the measured same-column probability settles below the value expected for a track with no diffusion at all (the dashed line in Figure 10). Both indicate that the charge undergoes some additional spreading, of order a micron, while drifting across the depletion region, which our model neglects. The ultraviolet photons of interest here are absorbed within a fraction of a micron of the back surface, where the diffusion is dominated by the field-free region, and the back-surface probabilities in Table 4 are measured directly rather than extrapolated from the fit, so this does not affect our results.

The cutouts of every track, the fits, and the list of frames searched are distributed with the source code of this article, along with the code to reproduce the fits.

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